SOC-5811 Week 14: Transformations

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- How do we relax our assumptions around additivity and linearity?

Transformations

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- How do we relax our assumptions around additivity and linearity?
  - Additivity: add interactions.
  - Linearity: transform variables.
- What does "relax" mean in this context?
  - Rule of thumb: Inferences made with simpler models rely on *more* assumptions.

$$\text{Happiness}_i = \beta_0 + \beta_1 \text{Income}_i$$

Say I want to use this model to infer whether people with 150k are happier, on average, than people with 50k.

What assumption does this inference rely on?

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What are "transformations?"

- We aren't talking about *shifting* or *rescaling* predictor variables, which changes our interpretation of the coefficients on those variables.
- We are talking about changing the *functional relationship* between a predictor variable and the outcome.

Transformations

- Log-transforming

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- Log-transforming
- Polynomials

Transformations

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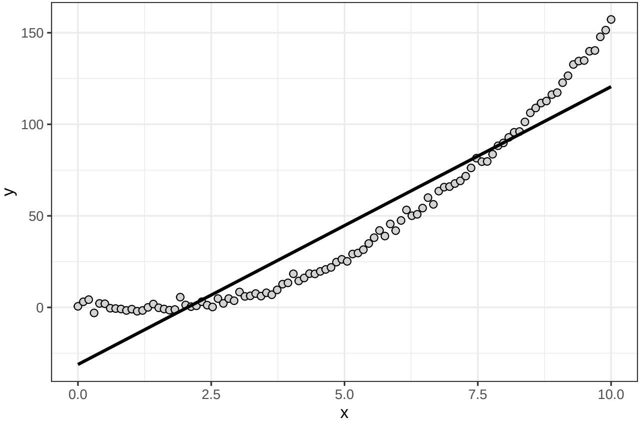
- Polynomials
- Splines

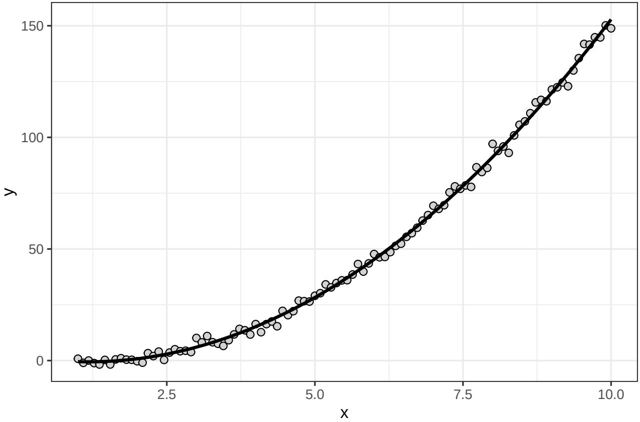
Transformations

$$y_i = \beta_0 + \beta_1 X_i$$

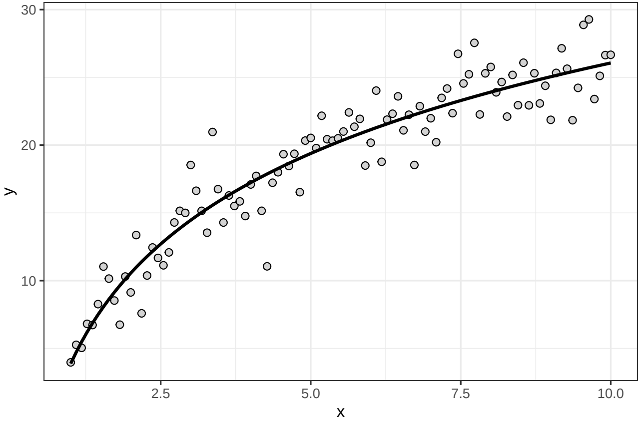


Transformations

$$y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2$$



$$y_i = \beta_0 + \beta_1 \log(X_i)$$



Transformations

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

```r
summary(mod)
```

```
##
## Call:
## lm(formula = y ~ log_x, data = d)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.8128 -1.1737  0.2431  0.9711  5.4042
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.8778     0.5343   7.258 9.36e-11 ***
## log_x         9.6341     0.3208  30.027  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.917 on 98 degrees of freedom
## Multiple R-squared:  0.902,  Adjusted R-squared:  0.901
## F-statistic: 901.6 on 1 and 98 DF,  p-value: < 2.2e-16
```

$$\beta_1 = 9.6$$

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

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Interpretation:

- A one-unit increase in ln(X) is associated with a 9.6 increase in Y.

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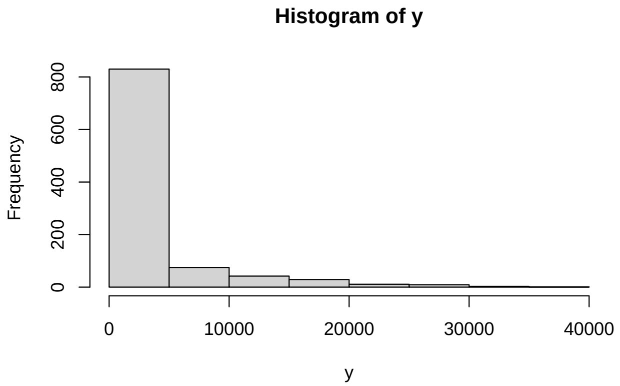
$$\beta_1 = 9.6$$

Interpretation:

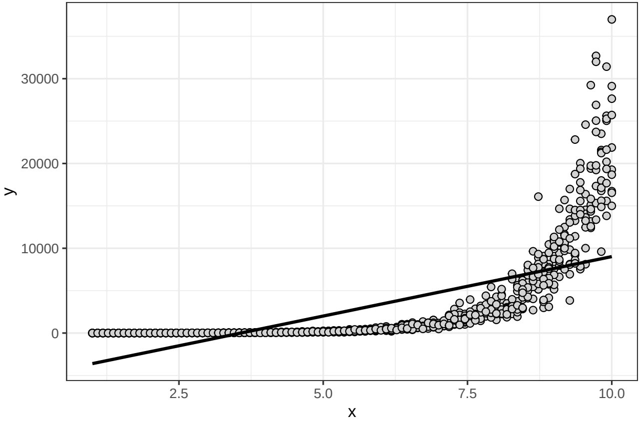
- A one-unit increase in ln(X) is associated with a 9.6 increase in Y.
- A 1% increase in X is associated with a .096 increase in Y.

Transformations



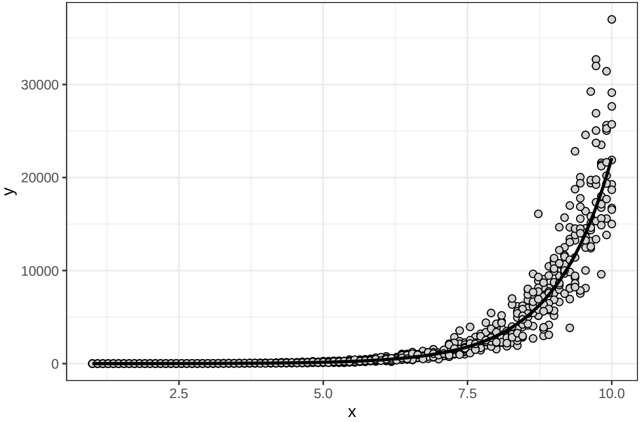
Transformations

$$y_i = \beta_0 + \beta_1(X_i)$$



Transformations

$$\log(y_i) = \beta_0 + \beta_1(X_i)$$



Why would we do this?

- Avoid negative predictions for an continuous outcome variable that is bounded at 0.

Why would we do this?

- Avoid negative predictions for an continuous outcome variable that is bounded at 0.
- If we think the relationship between X and Y is multiplicative rather than additive.

- **Only the dependent/response variable is log-transformed.** Exponentiate the coefficient. This gives the multiplicative factor for every one-unit increase in the independent variable.

Transformations

- **Only the dependent/response variable is log-transformed.** Exponentiate the coefficient. This gives the multiplicative factor for every one-unit increase in the independent variable.

- **Only independent/predictor variable(s) is log-transformed.** Divide the coefficient by 100. This tells us that a 1% increase in the independent variable increases (or decreases) the dependent variable by (coefficient/100) units.

Transformations

- **Both dependent/response variable and independent/predictor variable(s) are log-transformed.** Interpret the coefficient as the percent increase in the dependent variable for every 1% increase in the independent variable.

Transformations

- Log-transforming

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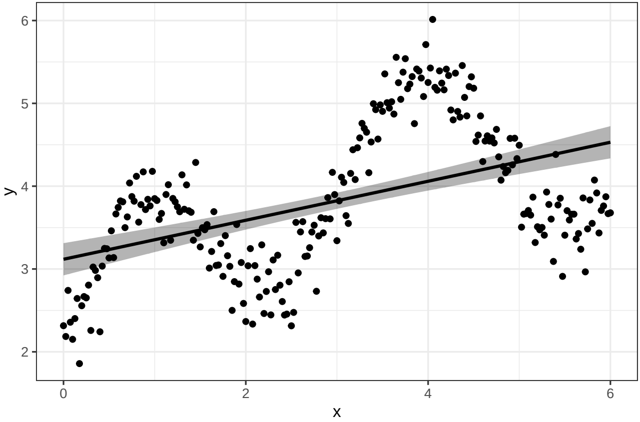
- **Splines**

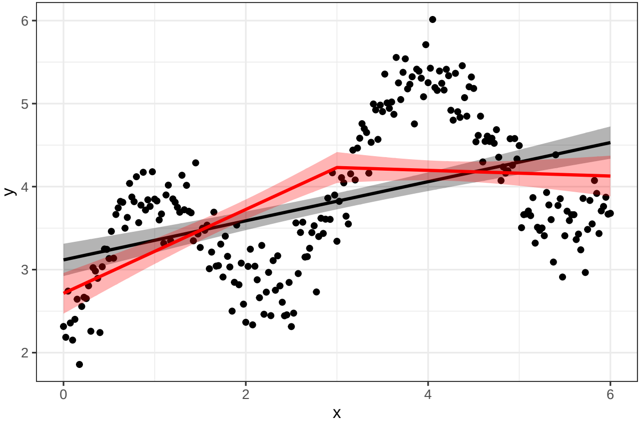
Transformations

```
mod <- lm(y~x, data=d)
```



Transformations

```
mod <- lm(y~lspline(x,knots=3), data=d)
```

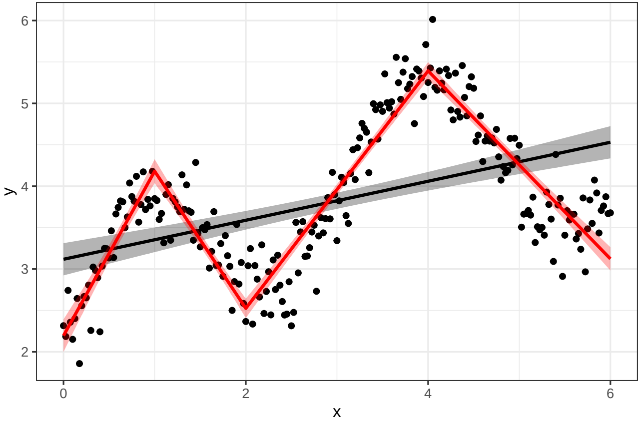


Transformations

```
##
## Call:
## lm(formula = y ~ lspline(x, knots = 3), data = d)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6634 -0.5753 -0.0236  0.6164  1.8192
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2.71471    0.12506  21.707  < 2e-16 ***
## lspline(x, knots = 3)1   0.50515    0.06110   8.268 9.63e-15 ***
## lspline(x, knots = 3)2  -0.03417    0.06110  -0.559    0.576
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7377 on 238 degrees of freedom
## Multiple R-squared:  0.293,  Adjusted R-squared:  0.2871
## F-statistic: 49.32 on 2 and 238 DF,  p-value: < 2.2e-16
```

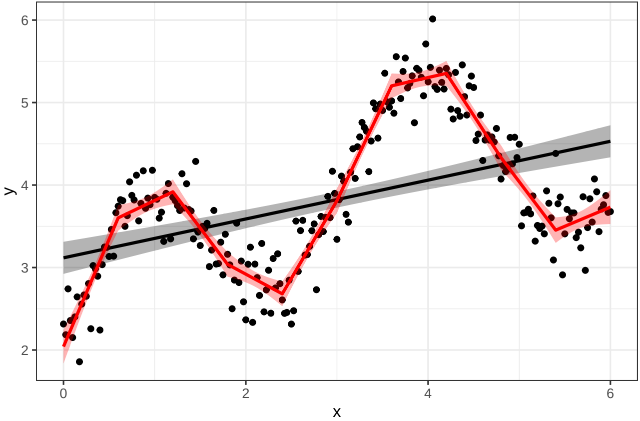
Transformations

```r
mod <- lm(y~lspline(x,knots=c(1,2,4)), data=d)
```



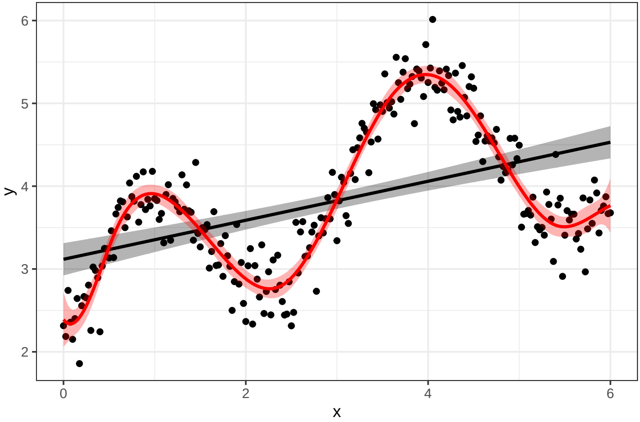
Transformations

```
mod <- lm(y~pspline(x,degree=1), data=d)
```



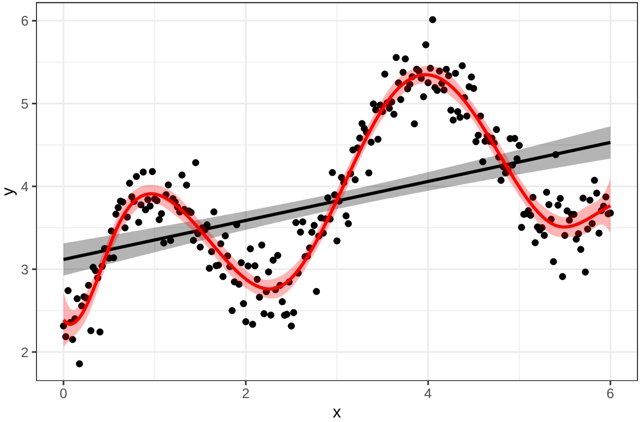
Transformations

```
mod <- lm(y~pspline(x,degree=3), data=d)
```



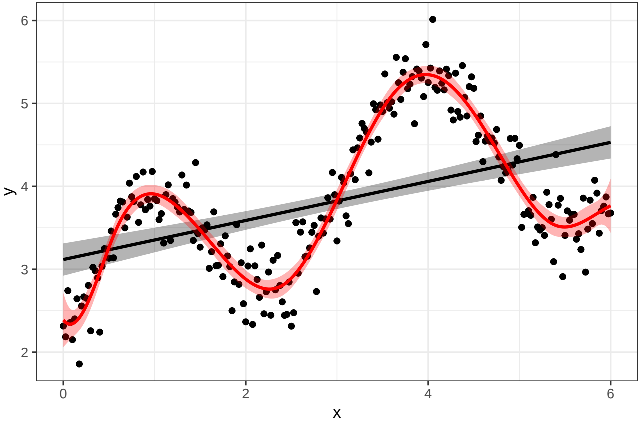
Transformations

Penalized cubic splines



Transformations

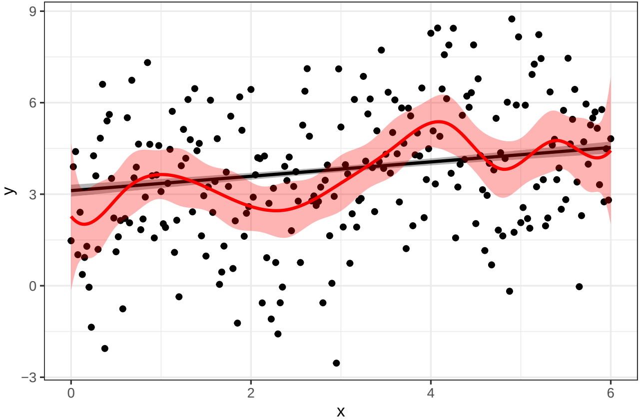
Which model do we like? Linear or penalized cubic spline?



Transformations

```
##
## Call:
## lm(formula = y ~ pspline(x, degree = 3), data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.72148 -0.16581  0.00908  0.17742  0.85489
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                5.7273     1.9864   2.883  0.00431 **
## pspline(x, degree = 3)1   -4.5816     2.3090  -1.984  0.04843 *
## pspline(x, degree = 3)2   -1.7045     1.8754  -0.909  0.36438
## pspline(x, degree = 3)3   -1.7892     2.0465  -0.874  0.38291
## pspline(x, degree = 3)4   -2.7016     1.9614  -1.377  0.16975
## pspline(x, degree = 3)5   -3.2271     2.0060  -1.609  0.10907
## pspline(x, degree = 3)6   -1.9271     1.9822  -0.972  0.33197
## pspline(x, degree = 3)7   -0.4043     1.9951  -0.203  0.83958
## pspline(x, degree = 3)8   -0.2632     1.9883  -0.132  0.89481
## pspline(x, degree = 3)9   -1.3715     1.9930  -0.688  0.49204
## pspline(x, degree = 3)10  -2.4898     1.9935  -1.249  0.21295
## pspline(x, degree = 3)11  -1.8631     2.0195  -0.923  0.35722
## pspline(x, degree = 3)12  -1.8242     2.8043  -0.651  0.51602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2807 on 228 degrees of freedom
## Multiple R-squared:  0.902,  Adjusted R-squared:  0.8968
## F-statistic: 174.8 on 12 and 228 DF,  p-value: < 2.2e-16
```

Transformations



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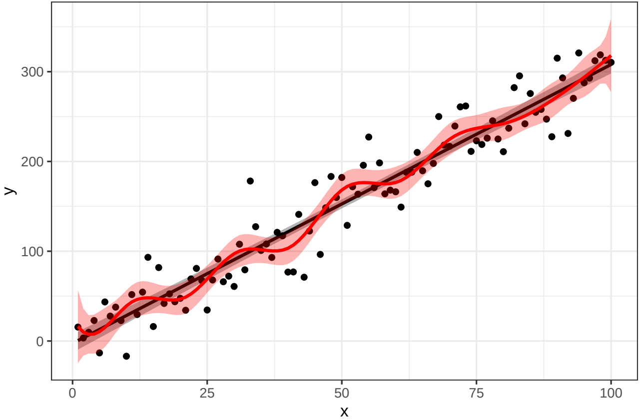
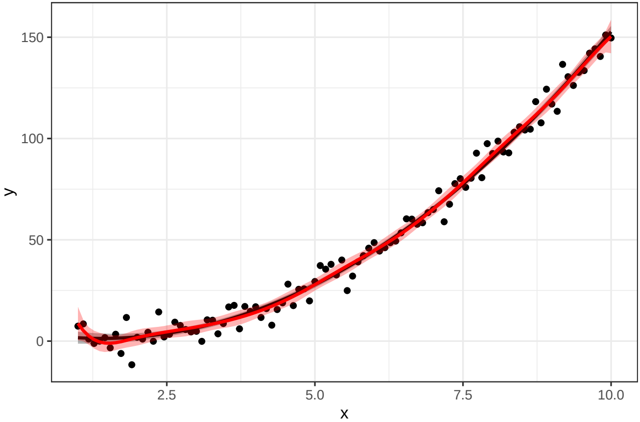


Transformations

- Splines vs. polynomials



Transformations

- **Quiz:** I'm interesting in the effect of X (i.e., β_1), and Z is a potential confounder. Inference on β_1 requires more assumptions in which of these models?

$$y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i$$

$$y_i = \beta_0 + \beta_1 X_i + \beta_2 S(Z_i)$$

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$$y_i = \beta_0 + \beta_1 X_i + \beta_2 S(Z_i)$$

- **Rule of thumb:** Inferences made with simpler models rely on *more* assumptions.

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- **Quiz:** If we want to relax our assumptions of additivity and linearity as much as possible, why not just use splines for all continuous variables and include all interactions?

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- **Quiz:** If we want to relax our assumptions of additivity and linearity as much as possible, why not just use splines for all continuous variables and include all interactions?
- There are models out there besides the conventional linear model that basically do this!
  - How do they handle the complexity vs. overfitting tradeoff? Cross-validation.

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  - But it is also the reason that has historically driven the use of simpler models; it is very easy to estimate the linear regression coefficient.
- We care about summarizing with average effects.
- We can demonstrate that our assumptions (e.g., linearity) do not substantively impact our conclusions.